

End-to-End Stock Market Analytics Dashboard

Python

Pandas

Plotly

Financial Analytics

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Project Objective

1. Analyze historical stock performance
2. Compare against market benchmark
3. Evaluate risk metrics
4. Generate automated insights
5. Export reusable dashboards/reports

Raw Market Data



Analytics Pipeline



Interactive Insights

Project Evolution

IBM Guided Project

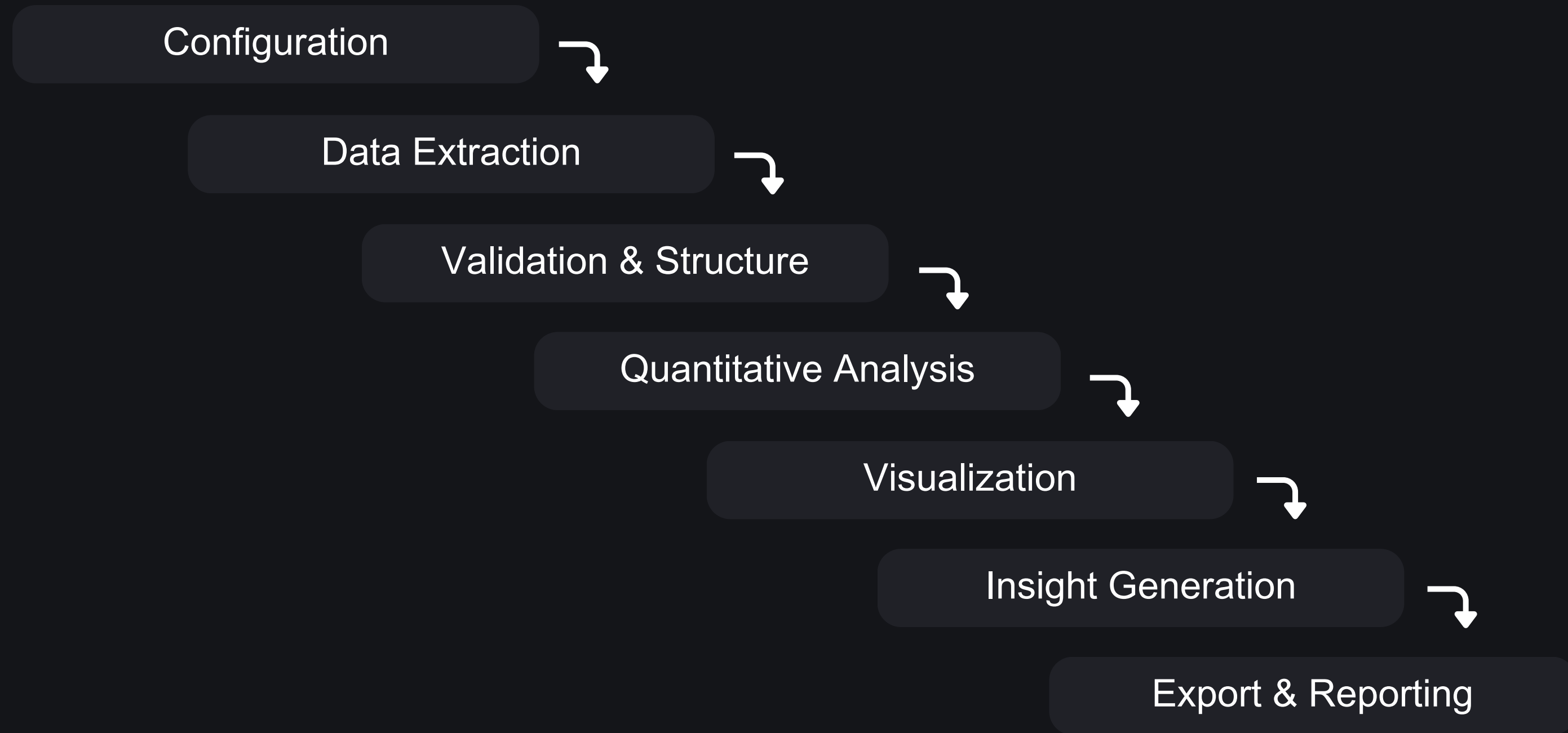
- Revenue scraping ✓
- Basic dashboard ✓
- Introductory analytics ✓



Expanded Analytics Pipeline

- Modular architecture ✓
- Financial metrics ✓
- Validation systems ✓
- Correlation analysis ✓
- Benchmark analysis ✓
- Automated insights ✓
- Export/reporting system ✓

Analytics Pipeline Architecture



Configuration Layer

Reusable:

Change settings in one place

Scalable:

Add stocks dynamically

Modular:

Pipeline references shared config

CONFIG

- tickers
- benchmark
- colors
- start_date
- end_date
- risk_free_rate

```
CONFIG = {  
    "tickers", "benchmark"...  
}
```

Data Extraction

`fetch_stock_data()`

- Downloads OHLCV data
- Caching system
- Cleans timestamps

`scrape_revenue()`

- Parses HTML tables
- Converts strings to numeric
- Structured quarterly revenue

yfinance API

**Web Scraping +
BeautifulSoup**



**Structured
Dataframes**



Date

Close



Validation & Structuring

TSLA
Dataframe



GME
Dataframe



AAPL
Dataframe



MSFT
Dataframe



Date	TSLA	GME	AAPL	MSFT
...	...	...	...	...
...	...	...	...	...
...	...	...	...	...

Missing values handled ✓

Date alignment ✓

Validation report ✓

```
pd.concat(frames, axis=1)
```


Quantitative Analysis Engine

Close Prices



Daily Returns



Risk & Returns Metrics

Total Return
Overall growth %

Volatility
Risk measurement

Sharpe Ratio
Risk measurement

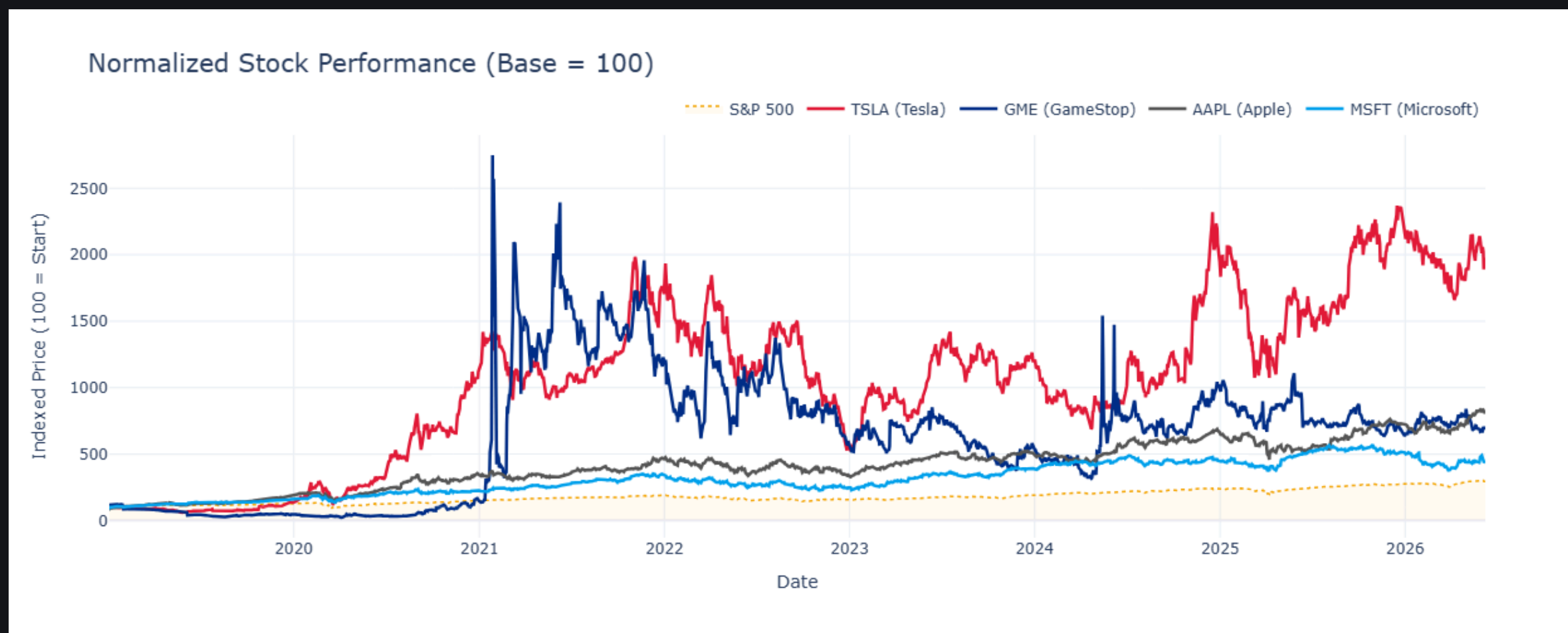
Max Drawdown
Largest decline

CAGR
Annualized growth

Core Calculations

```
daily_ret = prices.pct_change()  
ann_vol = daily_ret.std() * np.sqrt(252)  
drawdowns = (prices - cum_max) / cum_max
```

Normalized Performance



TSLA: +1,878% (10x the S&P 500)

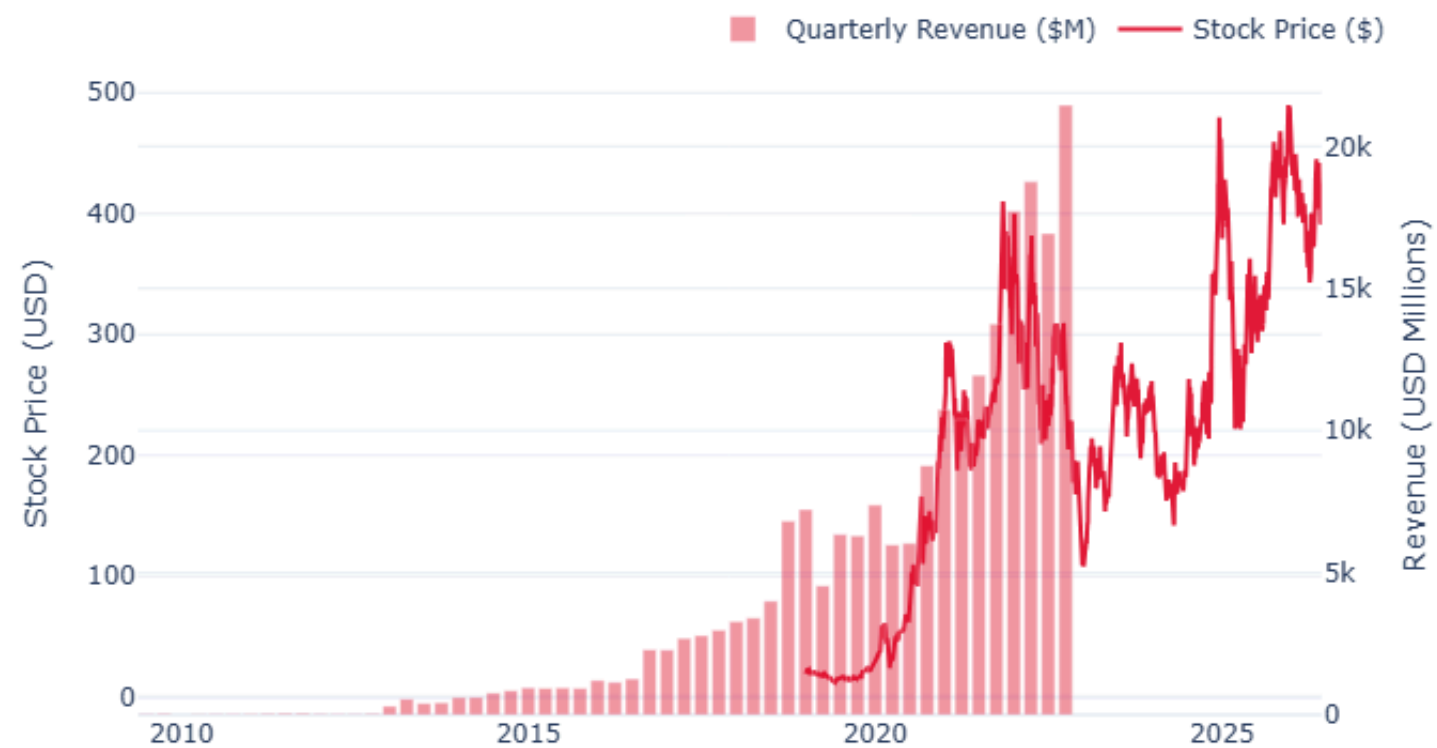
GME: January 2021 peak at 27x starting value

All four stocks outpaced the S&P 500 over the full period

```
rebased = close_matrix.div(close_matrix.iloc[0]) * 100
```

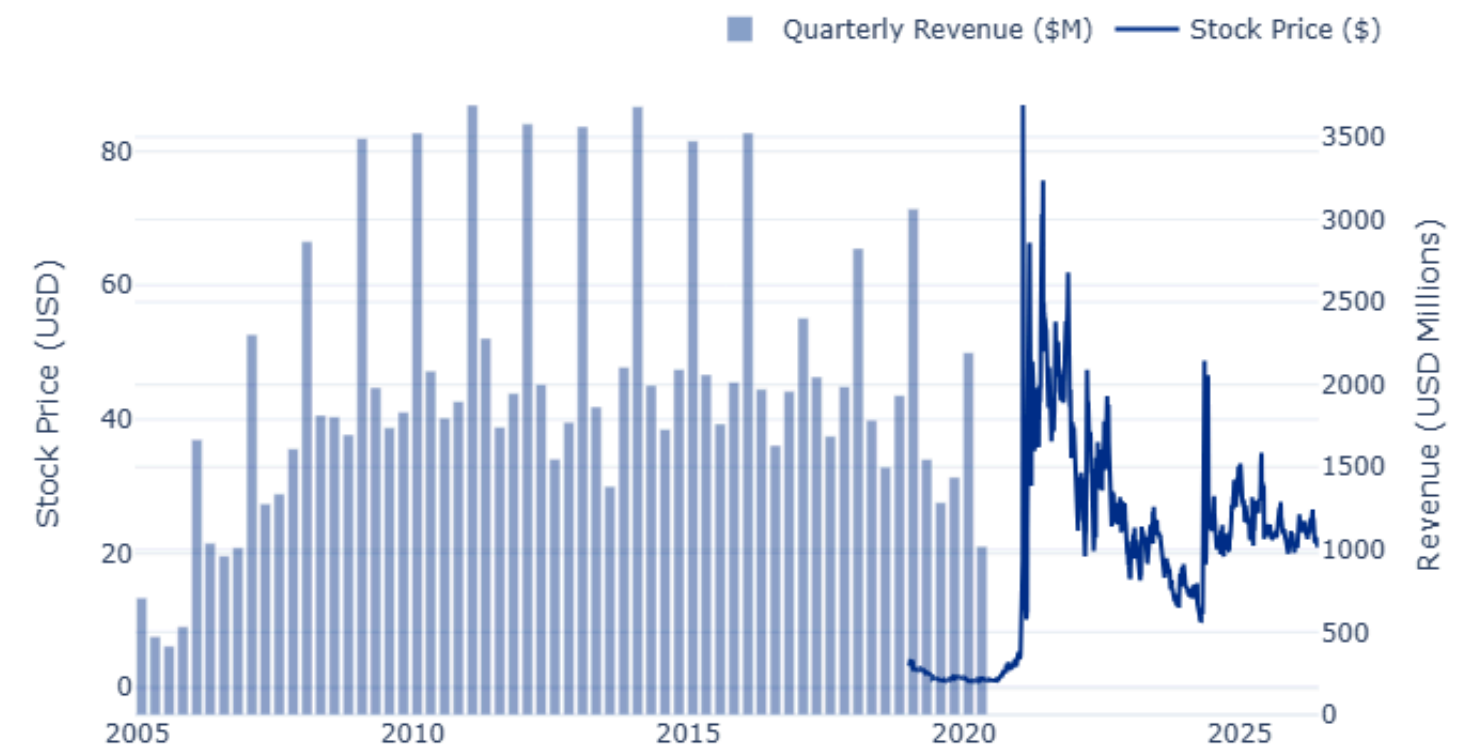
Revenue vs Stock Price

Tesla (TSLA) — Stock Price vs. Quarterly Revenue



Revenue data: IBM archive snapshot, through Q3 2022. Price data: yfinance, current.

GameStop (GME) — Stock Price vs. Quarterly Revenue



Revenue data: IBM archive snapshot, through Q1 2020. Price data: yfinance, current.

Question: Does revenue growth align with stock performance?

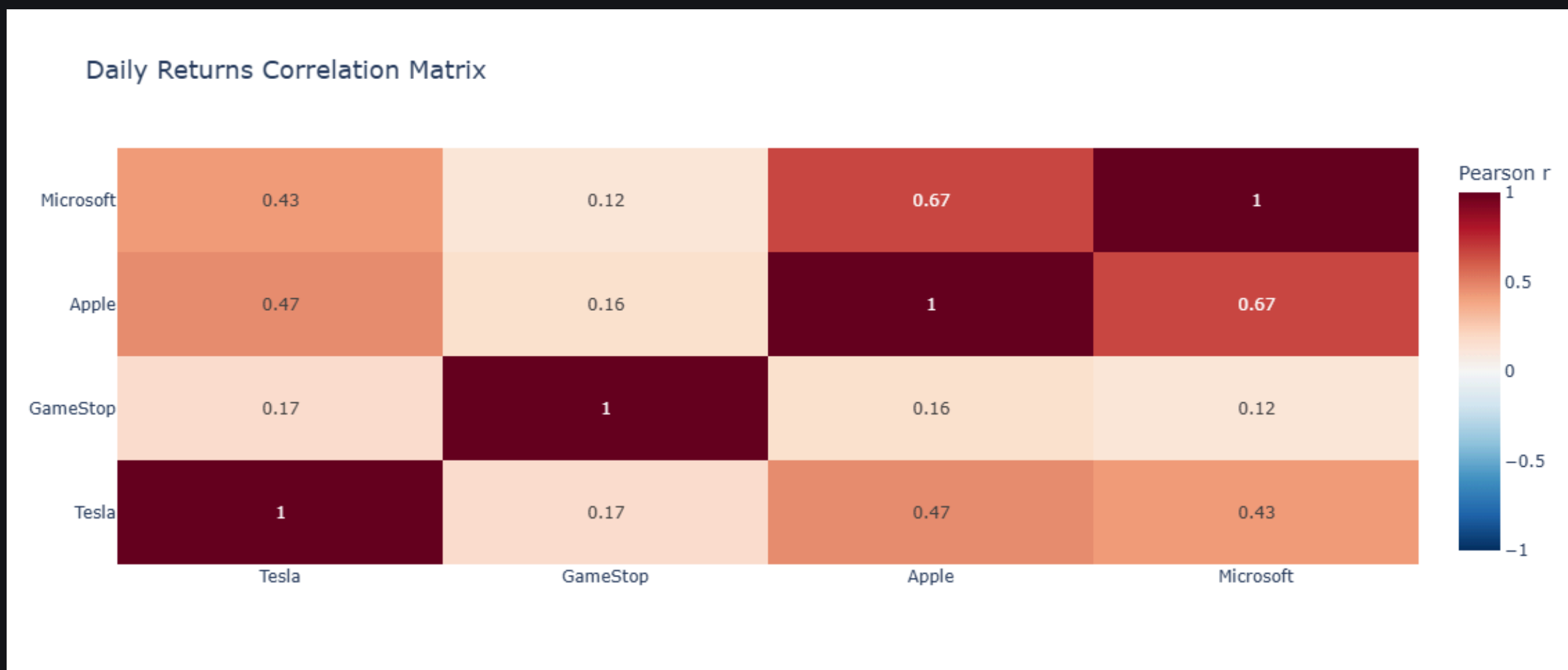
Moving Averages Volume - TSLA



Moving Averages Volume - GME



Correlation Analysis



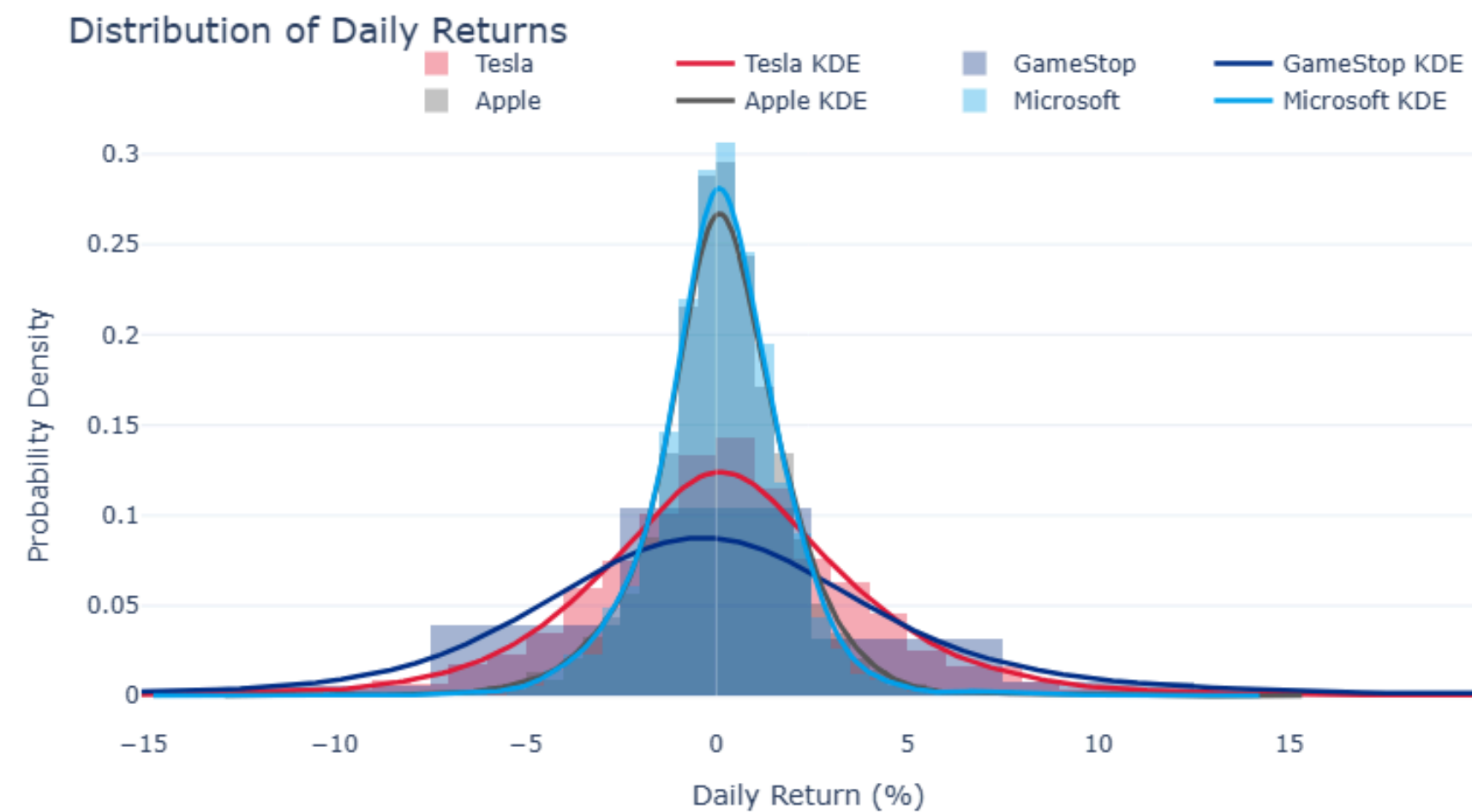
+1 = Highly related

0 = Unrelated

-1 = Inverse relationship

```
corr = returns.corr()
```

Return Distributions

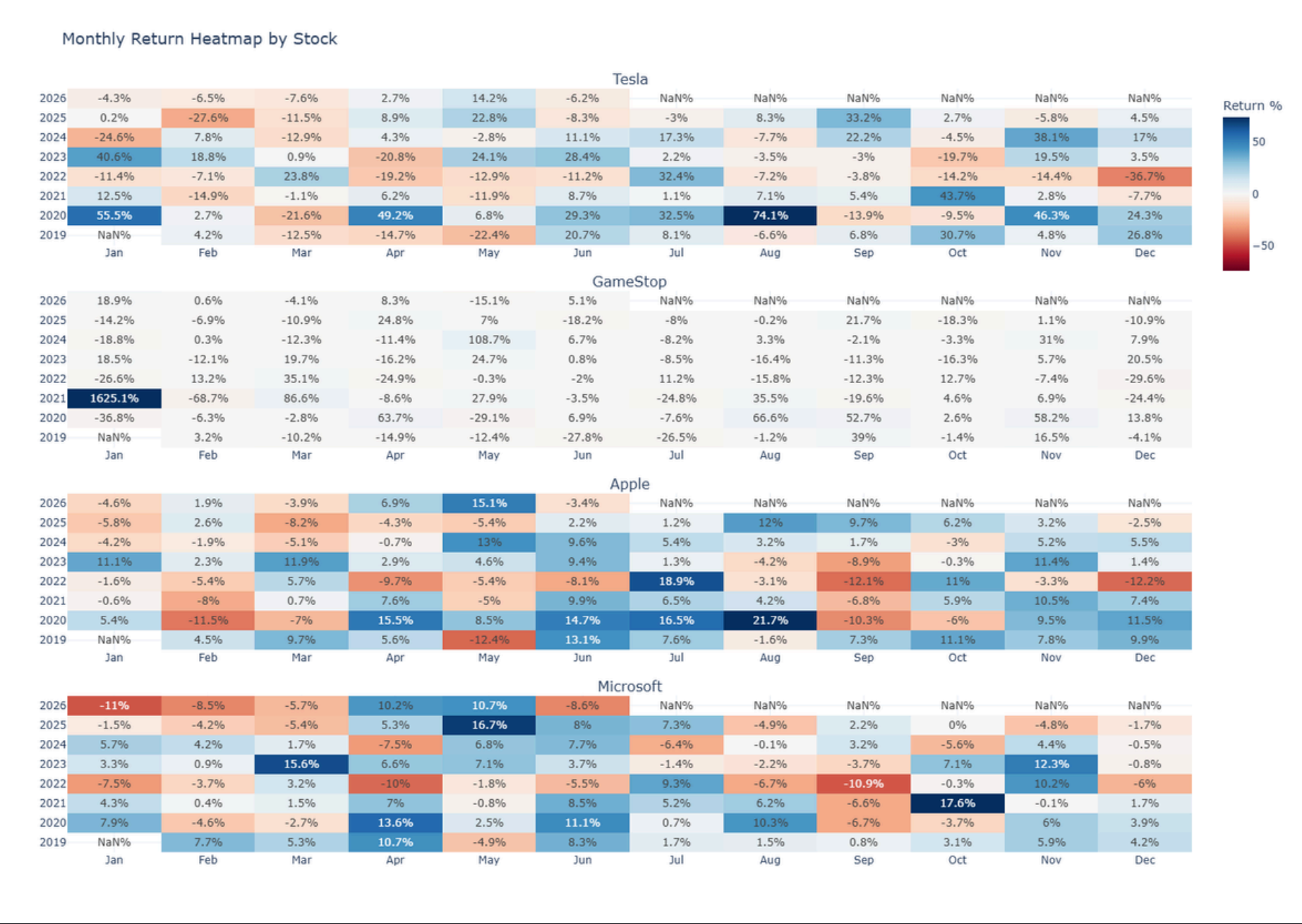


Center = Typical Returns

Spread = Volatility

Tails = Extreme Events

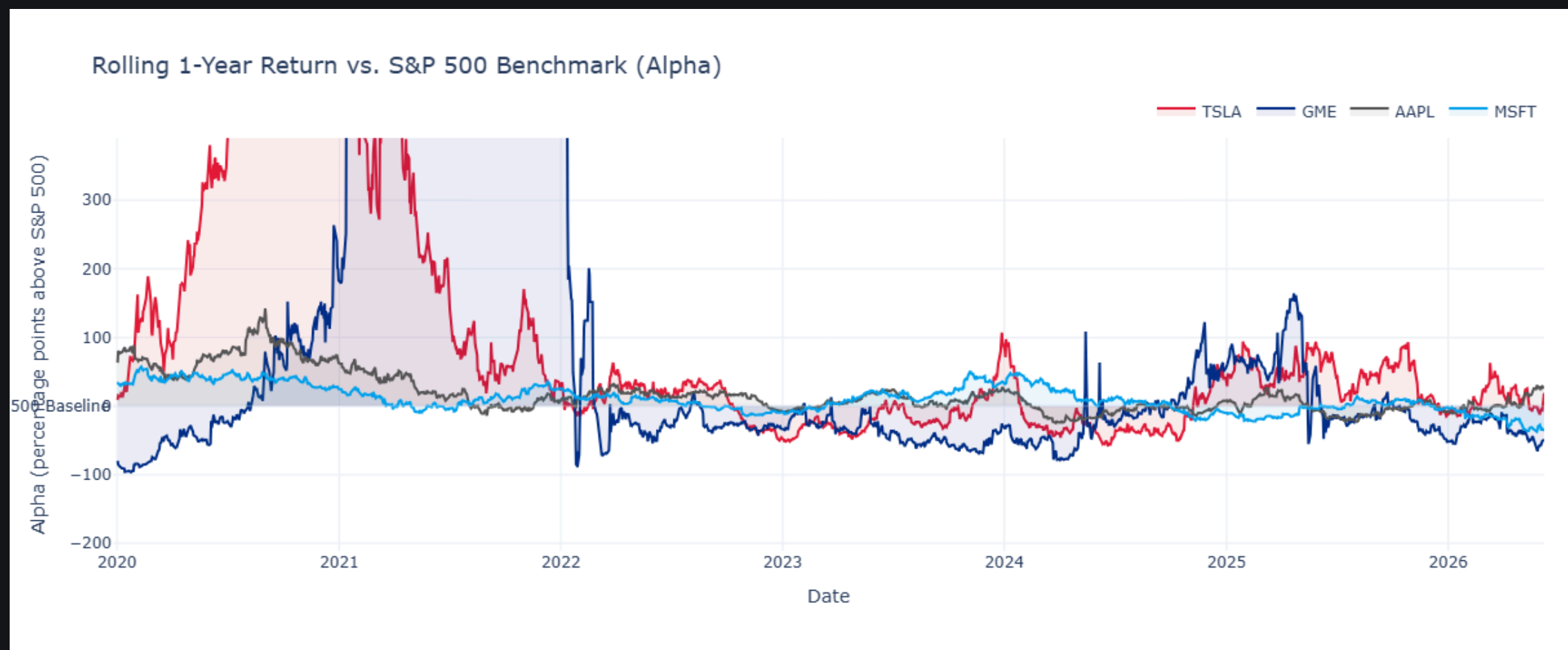
Monthly Return Heatmap



Blue = Strong Month

Red = Weak Month

Performance vs Market Benchmark



Above Zero =

Outperforming S&P 500

Below Zero =

Underperforming S&P 500

$$\text{alpha} = \text{stock_rolling} - \text{bench_rolling}$$

Automated Insights

PERFORMANCE: Tesla (TSLA) delivered the highest total return at +1878.0%, while Microsoft (MSFT) was the weakest at +337.1%.

RISK: GameStop (GME) had the highest annualized volatility at 134.6%, making it the riskiest holding in the group.

RISK-ADJUSTED RETURN: Apple (AAPL) showed the best risk-adjusted performance with a Sharpe Ratio of 1.01.

DRAWDOWN: GameStop (GME) experienced the steepest peak-to-trough decline of -88.5%, highlighting significant downside risk at certain points.

CORRELATION: Apple and Microsoft had the highest correlation (0.67), suggesting they respond similarly to market-wide events.

EVENT: GameStop surged 1,745% in January 2021 during the Reddit-driven short squeeze, briefly reaching \$86.88 before collapsing.

```
best = metrics_df["Total Return (%)"].idxmax()
insights.append(f"{best} delivered the highest return at {metrics_df.loc[best, 'Total Return (%)']:+.1f}%")
```

Exports

Data	metrics_summary.csv · all_close_prices.csv
Insights	insights.txt
Visuals	figures/ (6 interactive HTML dashboards)

Conclusion

Technical Skills

Python
Pandas
Plotly
Web Scraping

Analytics Skills

Financial Metrics
Correlation Analysis
Risk Analysis

Project Outcomes

Automated Insights
Interactive Dashboards
Exportable Reports